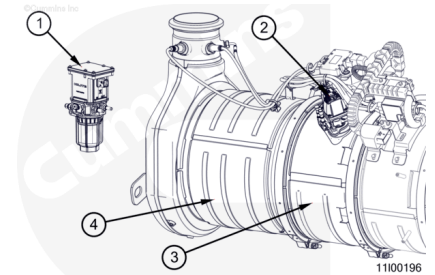


Trainings ▾

General Information

The selective catalytic reduction (SCR) system is used to decrease the nitrogen oxides (NOx) emissions from the vehicle tailpipe. The system is composed of several main components:

1. Aftertreatment diesel exhaust fluid (DEF) dosing unit
2. Aftertreatment DEF dosing valve
3. Aftertreatment decomposition tube
4. Aftertreatment SCR catalyst.



⚠ CAUTION ⚠

It is unlawful to tamper with, modify, or remove any component of the SCR system. It is also unlawful to use DEF that does not meet the specifications provided or to operate the vehicle/equipment with no DEF.

DEF is required for an engine equipped with an SCR system. DEF is a fluid that is sprayed into the exhaust gas prior to the aftertreatment SCR catalyst. The DEF vaporizes and decomposes to form carbon dioxide and ammonia. The ammonia reacts with the NOx emissions over the aftertreatment SCR catalyst to form nitrogen and water.

DEF:

- may have a slight ammonia smell
- is colorless
- is non-toxic and non-polluting
- is non-flammable.

Urea is naturally occurring and is biodegradable.

Note : Cummins Inc. supplies the aftertreatment DEF controller, aftertreatment dosing unit and aftertreatment DEF dosing valve, and the aftertreatment SCR catalyst. The vehicle manufacturer supplies the DEF tank, the DEF lines, the DEF tank temperature, the DEF level, and quality sensors.

The aftertreatment DEF dosing unit pumps DEF from the DEF tank to the aftertreatment DEF dosing valve. The aftertreatment DEF dosing unit is coolant heated and contains a filter that is a maintenance item.

The aftertreatment DEF dosing valve is DEF cooled and sprays DEF into the exhaust.

The engine control module (ECM) controls the amount of DEF sprayed into the exhaust. It also controls the DEF tank heater and DEF line heaters.

The aftertreatment SCR catalyst uses DEF to reduce the nitrogen oxide emissions in engine exhaust to nitrogen and water. The aftertreatment SCR catalyst itself requires no maintenance.

A vehicle with SCR will be equipped with an additional lamp on the dashboard, the aftertreatment DEF lamp. This lamp, along with the CHECK ENGINE lamp and STOP ENGINE lamp, alerts the operator to the level of the DEF tank. As the DEF tank level approaches empty, the aftertreatment DEF lamp will illuminate and engine power will be reduced. Attempting to operate the vehicle with no DEF in the tank will result in the vehicle speed being limited.

DEF is sprayed into the exhaust when the temperature in the aftertreatment SCR catalyst reaches approximately 250°C [482°F]. The amount of DEF consumed will differ from vehicle to vehicle, as DEF consumption depends on engine speed and load.

Even though DEF freezes at approximately -12°C [11°F], the SCR system is designed to be frozen and thawed. The DEF tank is heated by engine coolant, and the DEF lines and aftertreatment DEF dosing unit are electrically heated. No operator interaction is needed when operating in cold temperatures; heating and thawing are controlled automatically by the ECM.

After turning the keyswitch OFF on a vehicle with SCR, a pumping sound may be heard from underneath the vehicle. This sound is the aftertreatment DEF dosing unit recirculating DEF through the system and returning it to the tank. This is normal system operation. The recirculation process may take up to 15 minutes to complete. To avoid system damage, do **not** disconnect the vehicle batteries during this process.

Note : When the dosing valve is dosing DEF into the exhaust, there is a noticeable clicking sound. This is normal and does not indicate that the valve has malfunctioned.

Under certain conditions (cold or very dry), water condensation, in the form of water vapor, can be seen coming from the vehicle tailpipe. This is normal operation and will clear within a few minutes of normal vehicle operation.